MINNEMUDAC 2026 | GRADUATE DIVISION

Demand is What We See. Need is What We Miss.

Recorded food-shelf activity in Minnesota follows population and the network. Need, measured two different ways, barely moves it. Here is where the gap is, why it is there, and what August 2026 looks like.

Data Detectors · Valentino, Dhyan, Uddam · 17 October 2026

SIX QUESTIONS, SIX ANSWERS

What we found, before the detail

Q1 Need vs. activity

Counts agree; rates do not. Per resident, recorded activity does not track the food-insecurity rate. The mismatch is worst in Wilkin, Nicollet and Pope.

Q6 A measure of need

A falsifiable definition, an index of seven inputs that are not demand. It and Map the Meal Gap disagree on who is in need, and neither predicts activity.

Q4 Food prices

Prices explain little. The real patterns are a 20% seasonal swing (October peak) and one 20% step in March 2023 that prices cannot cause.

Q5 SNAP

Demand stepped up about 19% in March 2023 almost everywhere, but not more where more people were on SNAP. No evidence of substitution.

Q7 August 2026

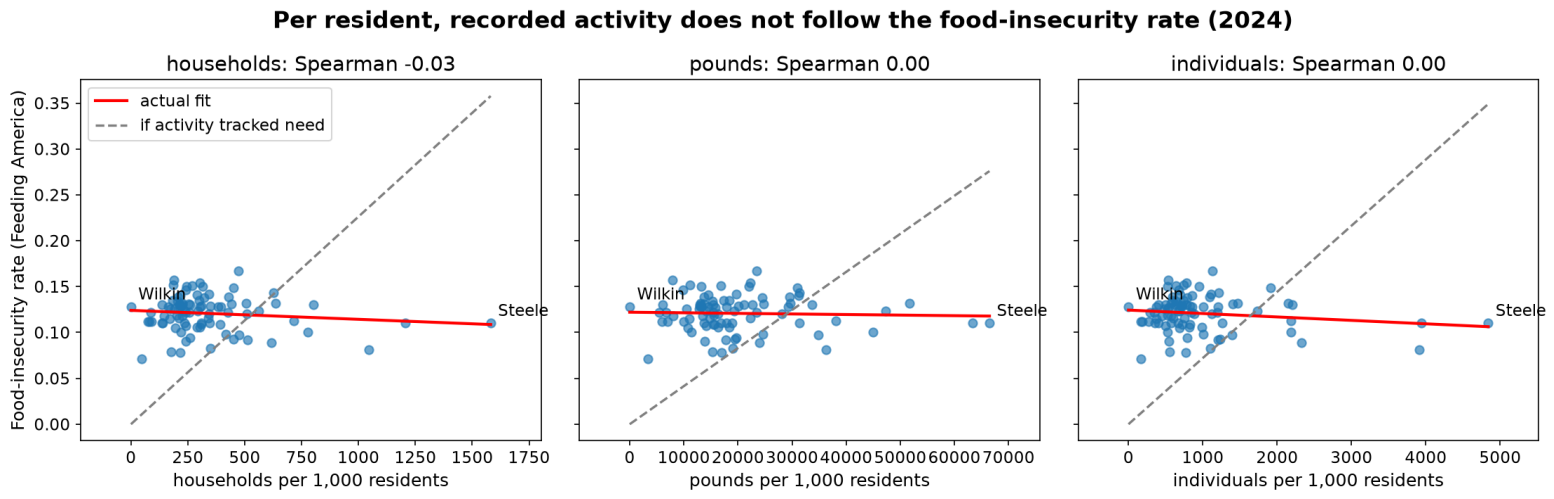
242,000 households, 12.9 million pounds, 746,000 individuals statewide. A pooled May-to-July ratio beat five other transparent methods on three backtest years.

Q8 Low-access areas

Food shelves reach urban low-access tracts as well as anywhere (98% within five miles) and rural ones worst of all (37%). The network repeats the retail gap.

Valentino

Counts agree with need. Rates do not.



Counts: tight, and steeper than 1:1

A county with 10× the food-insecure people records about 16× the activity.

Log-log slope 1.15 to 1.24, every interval above 1; R^2 0.84 to 0.88, every year, every metric.

Rates: nothing

Per 1,000 residents, Spearman with the food-insecurity rate is −0.03 to 0.17.

Pounds tracks need best, households worst, by a small, consistent margin of about 3 points of R^2.

Where it breaks down

Least activity for its need: Wilkin (no reporting food shelf), Nicollet, Pope, Lac qui Parle. **Most:** Steele, Washington, Rice.

Distance, rurality, no-vehicle and SNAP households explain 3% of the gap; none is clearly associated. The gap partly measures the network: counties with more sites record more per food-insecure person.

Sources: outputs/tables/q1_loglog_fits.csv, q1_correlations.csv, q1_residual_ranking.csv, q1_covariate_model.csv. Figure: 2024, 86 counties, Feeding America Map the Meal Gap 2026.

Define need before looking, then let the data argue

THE DEFINITION, WRITTEN ON 1 OCTOBER BEFORE THE INDEX EXISTED

A household is in need if, at any point in the past year, it worried about running out of food, regardless of whether it sought help.

The county index: seven inputs, equal weight

Households on SNAP · no vehicle · senior poverty · housing-cost burden · no diploma · low-income low-access share · debt in collections.

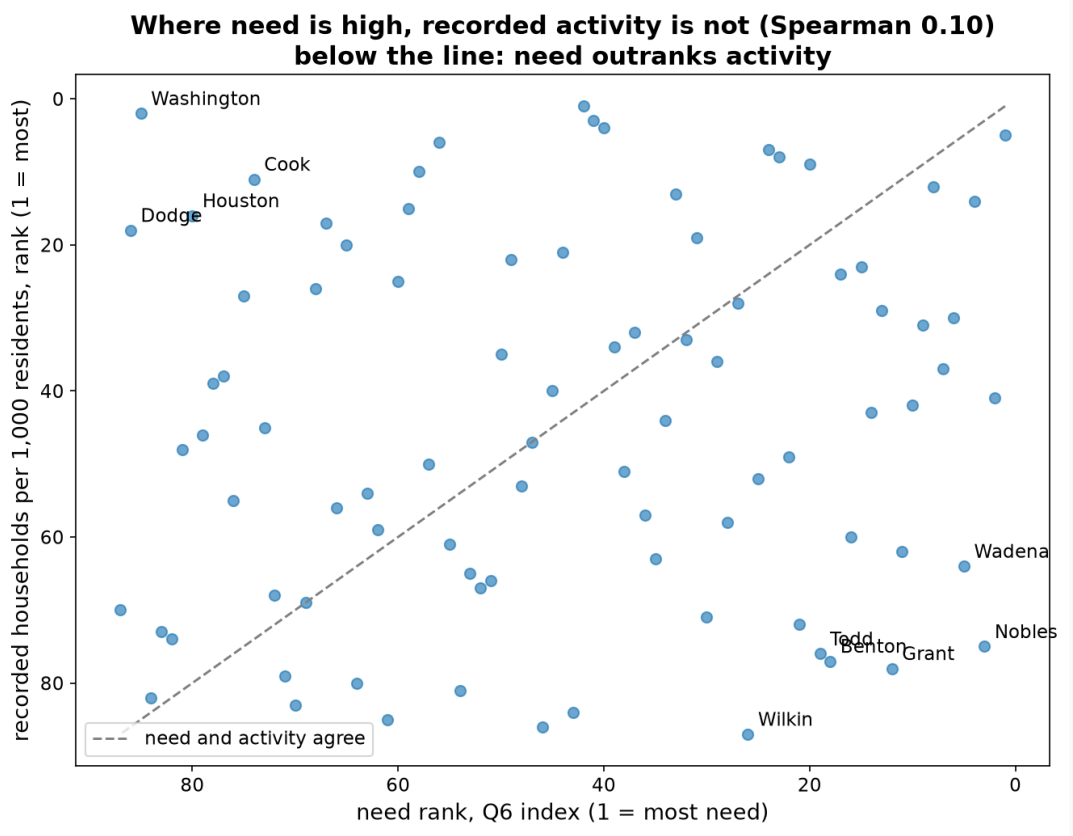
Each is measurable at zero demand and is not a Map the Meal Gap input. Barriers to reaching a food shelf (distance, rurality) stay out of the index and are used only to explain the gap.

What would prove us wrong

Reject: a household survey (USDA item 1) ranks counties differently.

Revise: the index collapses onto poverty ($R^2 \approx 0.9$), one input dominates, or the top ten changes when weights change.

Test passed: Wilkin was predicted near the 70th percentile of need; it landed at the 71st.

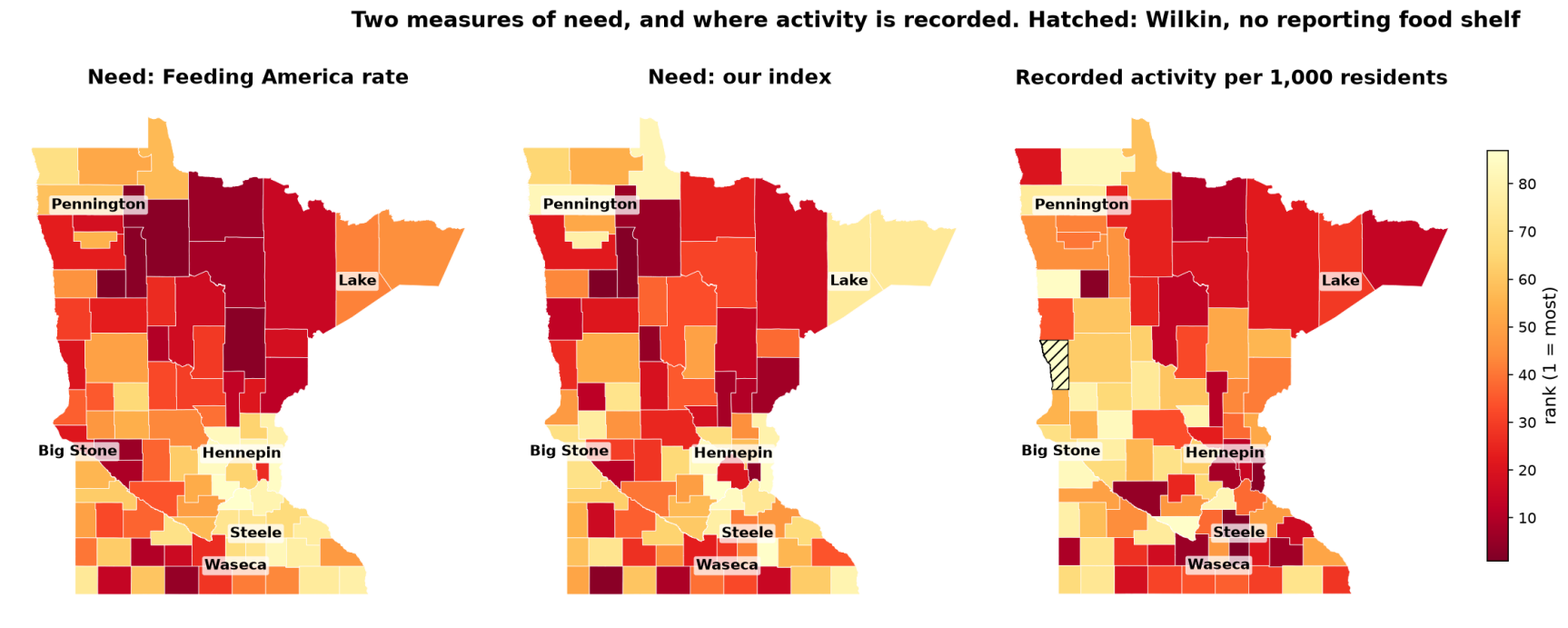


Need rank against activity rank, 86 counties. Spearman 0.10: knowing a county's need tells you almost nothing about how much activity it records. Below the line, need outranks activity: Nobles, Grant, Wilkin, Benton, Wadena. Above it: Washington, Dodge, Cook.

Sources: questions/q6_divergence_measure/addendum.md; outputs/tables/q6_need_index.csv, q6_divergence.csv, q6_stability.csv. Top seven on need hold under every alternative weighting.

Valentino

Two need measures agree; neither matches activity



MMG and the index: broadly the same

Spearman 0.77 between the two need rankings. They differ on five of the top ten: MMG finds need where incomes are low, the index where costs and barriers are high.

Neither predicts activity

County size alone explains 85% of recorded activity. Adding MMG or the index adds at most one point of R^2. Activity follows population, whichever way need is measured.

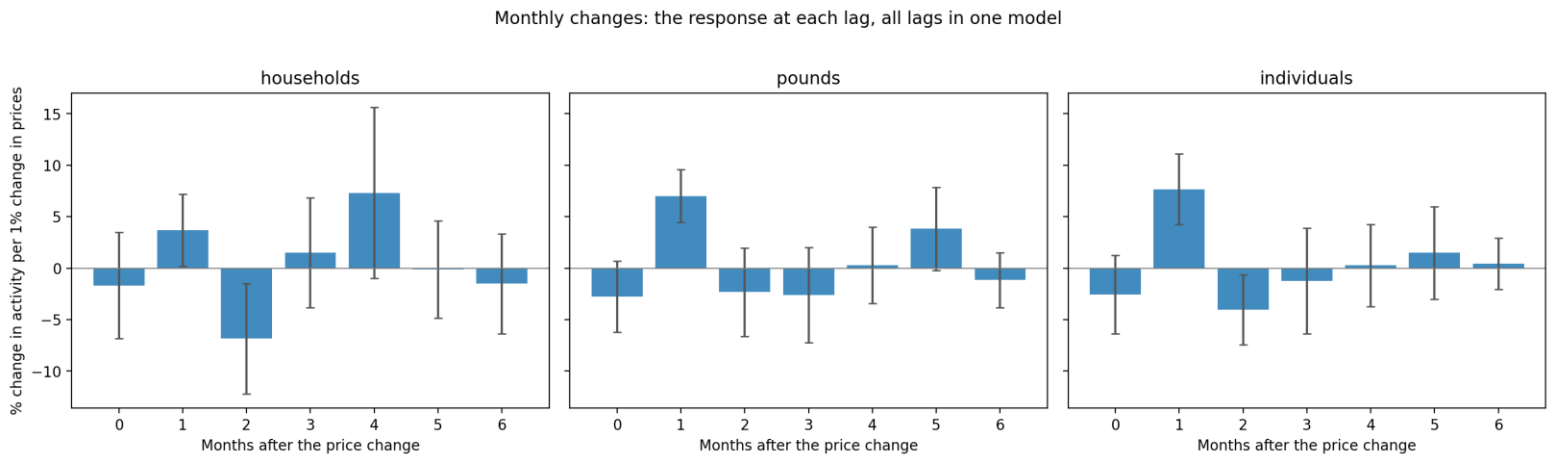
What explains the gap: barriers

No internet at home +0.41 per SD (p = 0.005) and **distance to a site** +0.23 (p = 0.025). Together with rurality and age, 17% of the divergence.

Sources: outputs/tables/q6_vs_mmg.csv, q6_barriers_model.csv (model 'with three more barriers'), q6_barrier_sides.csv. Maps: mean rank 2022 to 2024; hatched = Wilkin, no reporting food shelf.

Dhyan

Prices explain little; the calendar explains a lot



What prices do not explain: the calendar

October is the busiest month and January or February the quietest, in all three metrics: a swing of about 20%, the same every year. That is the pattern a food shelf can plan around.

Pounds per household falls every year (62 in 2022, 50 in 2025): the network spreads less food over more households, so pounds responds to supply, not only to demand.

Timing: about one month, if at all

Where a response shows, it is one month after the price change, in pounds and individuals, not households.

With SNAP and a trend in the model the price effect drops by a third.

The 2023 step is not a price story

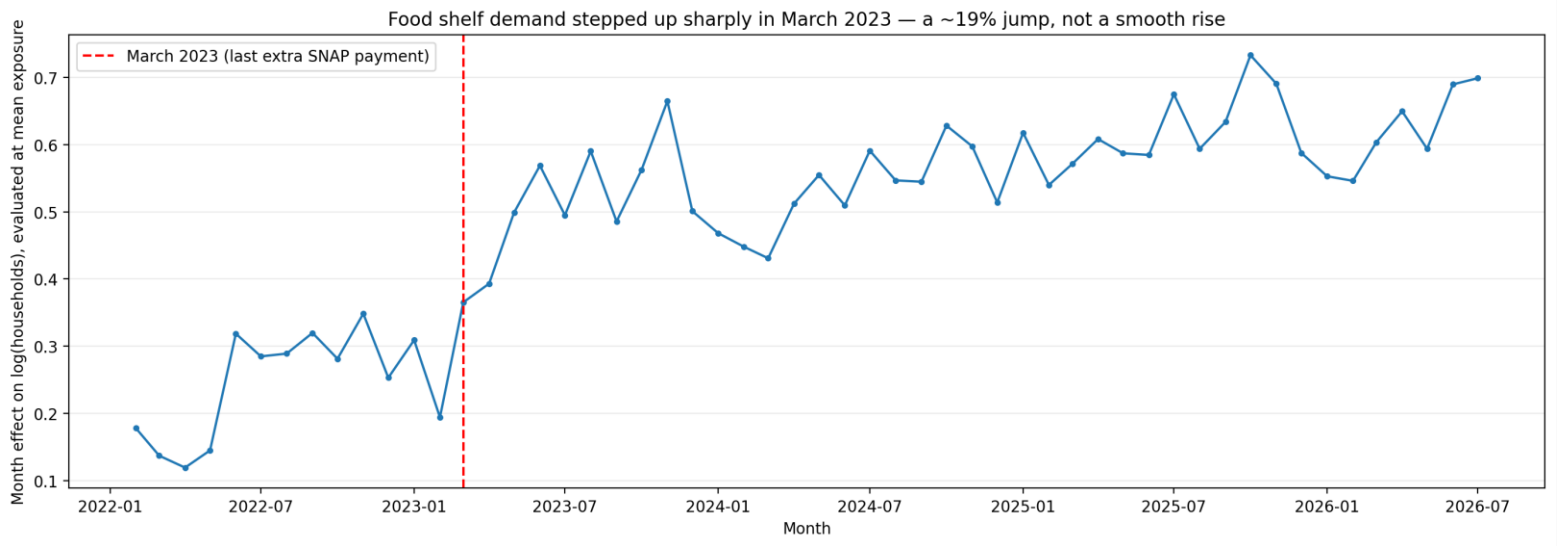
Activity grew most during the 2022-23 food inflation, but the largest single move is a 20% jump in March 2023: all three metrics, one month.

Prices fell smoothly through 2023 and cannot produce a step.

Sources: Dhyan, notebooks/q4 (dhyan_branch) and outputs/figures/q4_monthly_distributed_lag.png; BLS CPI CUUR0200SAF11 (food at home, Midwest, monthly). TODO Dhyan: confirm the lag-1 coefficient and regenerate the figure from analysis.py.

Dhyan

March 2023: a 19% step, not larger where SNAP was



The result

Exposure does not predict a larger rise: coefficients 0.013 and −0.009 (p = 0.42 and 0.33), metro and non-metro alike.

Demand stepped up about 19% in March 2023 in 91% of units (median +18%), regardless of how many of their people were on SNAP. No evidence that food shelves substitute for SNAP; demand rose through 2023 everywhere.

The event

Minnesota's last extra pandemic SNAP payment was issued in March 2023; households received the regular benefit only from April.

A dated, national cut, with a different bite in every county.

The design

Event study on 79 SNAP reporting units: unit and month fixed effects, errors clustered on unit.

Two exposure measures (share of people on SNAP before the cut), metro and non-metro separately.

Sources: Dhyan, notebooks/q5 (dhyan_branch), outputs/figures/q5_month_effects_event_study.png; MN DHS county SNAP caseloads (monthly, 79 reporting units). TODO Dhyan: numbers from analysis.py.

What to expect in August 2026, statewide

242,106
households

12.86 M
pounds

746,407
individuals

264,116
children

394,811
adults

87,480
seniors

87 county forecasts, five outcomes each, with 80% intervals from backtest errors (about −22% to +13% for Hennepin households). Hennepin alone: 72,769 households. Wilkin has no reporting food shelf and is imputed from rural peers, labelled as such.

The method, and why

Six transparent candidates backtested on August 2023, 2024 and 2025, each fold using only earlier months.

Winner: pooled May-to-July ratio. Average a county's last three months, scale by its historical August-to-quarter ratio, shrunk toward the statewide ratio. Lowest RMSE on all five scored outcomes.

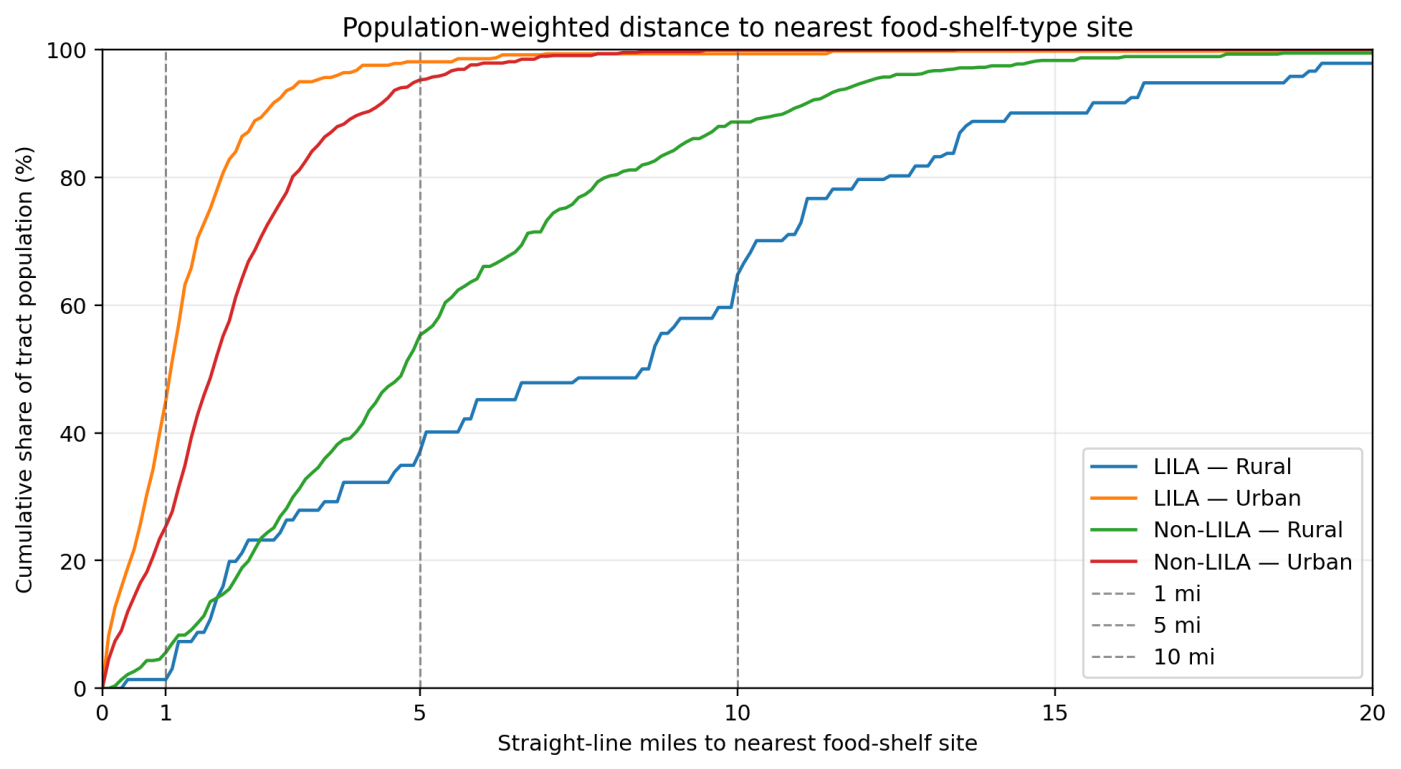
Accuracy and the leakage rule

County-level sMAPE in backtests: 12% (households, adults) to 17% (children).

The panel ends July 2026. An independent audit (19 checks, run by a non-author) confirms nothing dated August 2026 enters any input or any backtest fold.

Sources: outputs/prediction/Graduate_Predictions_Submit.csv (column sums), outputs/tables/q7_method_selection.csv, q7_forecasts.csv, questions/q7_forecast_august/LEAKAGE_AUDIT.md.

Fills the gap in cities, repeats it in rural areas



SHARE OF PEOPLE WITHIN 5 MILES OF A FOOD SHELF

98%
urban, low-access (LILA)

95%
urban, other

37%
rural, low-access (LILA)

55%
rural, other

Fill or repeat?

In rural low-access tracts the nearest supermarket averages 9 miles and the nearest food shelf 8: the two networks sit in the same places. Metro-adjacent and remote rural counties look alike (34% and 39% within five miles).

Limitations carried as numbers, not apologies: 2019 USDA flags moved to 2020 tracts by land-area overlap (23 of 1,505 tracts ambiguous); sites placed at tract interior points (median displacement bound 3.1 miles); 402 sites in this definition.

Sources: outputs/tables/q8_population_weighted_coverage.csv, q8_retail_pattern_comparison.csv (OSM supermarkets), q8_rural_adjacency_access.csv. USDA Food Access Research Atlas 2019.

THE THROUGH-LINE

Recorded activity follows population and sites. Need, measured either way, barely moves it.

Where it is missing

Far from a site and offline: Nobles, Grant, Wilkin, Benton, Wadena, and rural low-access tracts statewide.

What moves it over time

The calendar (October peak) and one step in March 2023, not prices and not SNAP exposure.

What to plan for

August 2026: 242,000 households and 12.9 million pounds, county by county, intervals included.

Every number traces to a script in github.com/dhyan95/MinneMUDAC. Questions welcome.

APPENDIX

Public data sources and why each one

Source	Exact series	Grain	Why this one	Used by
Feeding America	Map the Meal Gap 2025 (2019-23) and 2026 (2024)	county · annual	Mandated by the brief; the only county food-insecurity estimate	Q1, Q6, Q7
Census ACS 5-year	B22003, B08201, B17001, B25070/B25091, B15003, B16002, B28002, B05001	county · 5-yr	Only source for vehicle, housing-burden, internet and education at county grain	Q6, Q1
USDA ERS	Food Access Research Atlas (LILA flags); RUCC 2023	tract · 2019; county	The official low-access definition; RUCC is inside MMG	Q8, Q6, Q1
Urban Institute	Debt in America: share with debt in collections	county · 2025	Financial stress not captured by income; caveat: credit-file population only	Q6
BLS	CPI CUUR0200SAF11, food at home, Midwest	region · monthly	Finest monthly food-price series covering all of Minnesota	Q4
MN DHS	SNAP caseloads by reporting unit	unit · monthly	The only county-grain monthly SNAP series; publication lag noted	Q5
Census	County adjacency file 2023; tract relationship file 2010-2020; Gazetteer	county / tract	Spatial tests (Q1), crosswalk of 2019 flags to 2020 tracts (Q8)	Q1, Q8
USDA FNS / OSM	SNAP retailer locator (to 2025-12-31); Geofabrik MN supermarkets	point	Two independent store sets for the retail-pattern comparison	Q8

Full registry with publication lags and file paths: docs/DATA_SOURCES.md. Nothing dated August 2026 or later enters the repository.

APPENDIX

Methods, one line each

Q1 Log-log OLS of activity on food-insecure persons per county-year (2022-24); residual ranking; covariate model with clustered errors; Moran's I (0.06, p = 0.37), join counts and pooling tests for spatial pattern.

Q6 Within-year z-scores of seven inputs, equal weights; divergence = need rank minus activity rank; stability under 10 alternative weightings and random (Dirichlet) weights; barrier regression; nested models against MMG.

Q4 Year-over-year changes, distributed lag 0 to 6 months with month-of-year effects; SNAP and trend as controls; residual seasonality.

Q5 Event study on 79 SNAP reporting units: unit and month fixed effects, exposure × post interaction, clustered errors, metro split.

Q7 Six baselines backtested on August 2023-25; pooled May-July ratio shrunk toward the state; 80% intervals from backtest residual quantiles; independent 19-check leakage audit.

Q8 Area-weighted 2010→2020 tract crosswalk; population-weighted distance bands to nearest site; comparison with supermarket distances; urban/rural and adjacency splits; displacement bounds for site locations.